Lokesh Venkatesh

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Research Interests: **Theory of Complex Systems** | **Computational Biology** | **Social Network Dynamics**

Education

Indian Institute of Science Education and Research (IISER), Pune — Nov '22 — Present
Integrated BS–MS in Natural Science *(Expected: May '27)* — CGPA: 7.8

Experience

Master's Thesis Project Intern, MPIPZ Cologne — Jun '26 — Present
Advisors: Dr. Pau Formosa-Jordan, MPIPZ & Dr. Josep Mercadal Melia, MPIPZ

- Carrying out my final year **Master's Thesis project** in the Formosa-Jordan group at the Max Planck Institute for Plant-Breeding Research.
- Project title: **Quantitative analysis of pattern formation in the context of Trichome distributions in *Arabidopsis***
- I am using theoretical frameworks from dynamical systems theory to model pattern formation in the context of trichome distributions on the growing *Arabidopsis* leaf.

Summer Intern at the Chronobiology Lab, NCBS Bangalore — May '25 — Jul '25
Advisor: Dr. Shaon Chakrabarti, NCBS Bengaluru

- Project title: **Self-supervised Representation Learning for Circadian Time Series Datasets**
- Implemented representation learning algorithms on patient actigraphy datasets, followed by clustering, in order to obtain their learned low-dimensional representations and isolate what specific variables carry the most significance in the multivariate time series.

Semester Project Student at the COSPI Lab, IISER Pune — Aug '24 — Apr '25
Advisor: Prof. MS Madhusudhan, IISER Pune

- Project title: **Tracing the evolution of the Mitochondrial Genome using Motif Distributions**
- Analysed motif distributions across a database of mitochondrial genomes as a metric to reconstruct phylogenetic relationships over the course of its evolution, as well as identify genomic translocations, and look at possible regulatory correlations between mitochondrial & nuclear genome promoter sequences.

Academic

Schools & Workshops

Spins, Games & Networks, 2025 — Dec '25
Two-week fully-funded winter school on theoretical approaches to modelling complex systems at IMSc Chennai

NeuroAI Virtual Summer School — Jul '25
Ten-day interactive online course on principles connecting artificial & biological neural nets by Neuromatch Academy

Brain, Dynamics & Computation, 2025 — May '25
Two-week fully-funded summer school on network science-approaches to comp. neuroscience at IMSc Chennai

Course & Reading Projects

Theory and modelling of Complex Systems — Reading Project — Sep '25 — Jan '26
Undertook a reading project during the semester to explore existing theoretical frameworks & tools used to model complex biological and social systems.

Relevant Coursework

Data Science — Applied Mathematical Methods, Image & Video Processing, Signal Processing, Deep Neural Networks, Mathematical & Statistical Optimisation, Computational & Functional Genomics

Physics — Mathematical Methods for Physicists, Classical Mechanics, Statistical Mechanics, Nonlinear Dynamics, Physics of Biological Systems

Mathematics — Real Analysis, Linear Algebra, Introductory Probability Theory, Graph Theory, Markov Chains, Mathematical Finance

Biology — Systems Biology, Neurobiology, Bioinformatics, Cellular Biophysics I & II, Animal Behaviour, Ecology & Evolution, Cell Biology, Physiology, Genetics

Humanities & Social Sciences — Science & the World, Philosophy of Science, Cognitive Bases of Science, Science & Society, Public Policy I & II

Skills

Computational: Numerical Simulations, Bifurcation Analysis, Deep Neural Networks, Representation Learning Algorithms, Bioinformatics

Programming: Python (NumPy, Pandas, SciPy, PyTorch, Scikit-learn), (BifurcationKit, DynamicalSystems), Scripting; (Markdown, LaTeX)

Languages: **English**, **Tamil** (bilingual); **Hindi** (working proficiency); **German** (elementary proficiency)

Scholarships & Awards

INSPIRE's Scholarship for Higher Education (SHE)
INSPIRE-SHE (4 Lakh INR $\approx 3,700$ EUR) - awarded to 0.1% of students nationwide by the Department of Science and Technology, Government of India

Extracurricular

Academic Support Group, IISER Pune Aug '23 — Sep '24
Senior teaching & logistics volunteer for a student collective mentoring socio-economically underprivileged students

Kalpa — IISER Pune's Student Media House Mar '23 — Aug '24
Member of the student media body's core team that coordinated investigative journalism projects on campus affairs

References

Dr. Pau Formosa-Jordan (MPIPZ Cologne)
Email: PFORMOSA@MPIPZ.MPG.DE

Prof. M.S. Madhusudhan (IISER Pune)
Email: MADHUSUDHAN@IISERPUNE.AC.IN